THE HALF-KERNEL OF THE ZETA SPECTRAL PROCESS

1. Setup

Let θ be the Riemann–Siegel theta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $u = \theta(t)$ the phase variable. Define the time-changed process

$$h(u) := \frac{Z(t(u))}{\sqrt{\theta'(t(u))}}, \qquad t(u) = \theta^{-1}(u),$$

which is weakly stationary on $[\theta(T_0), \infty)$ with $T_0 = 200$. Its Cramér spectral representation is

$$h(u) = \int_{-1}^{1} e^{i\nu u}\, dM(\nu), \qquad \mathbb{E}|dM(\nu)|^2 = S(\nu)\, d\nu + w\, \delta_{-1},$$

where $\operatorname{supp} S \subseteq [-1, 1]$ and $S(\nu)\, d\nu$ is the absolutely continuous part of the spectral measure ν.

The spectral density S is given by the limit

$$S(\nu) = \lim_{T\to\infty} |K_T(\nu)|^2, \qquad K_T(\nu) = \frac{1}{2\pi} \int_{U_0}^{U_T} h(u)\, e^{-i\nu u}\, du, \tag{1}$$

which in terms of the Riemann–Siegel expansion

$$h(u) = \sum_{n=1}^{N(t(u))} \frac{e^{i(u - t(u)\log n)} + e^{-i(u - t(u)\log n)}}{\sqrt{n}\,\sqrt{\theta'(t(u))}} + \text{lower order}$$

takes the form

$$S(\nu) = \lim_{T\to\infty} \frac{1}{4\pi^2} \sum_{m,n=1}^{N_T} \frac{1}{\sqrt{mn}} \iint \frac{e^{i\Psi_m(s) - i\Psi_n(r)}}{\sqrt{\theta'(t(s))\,\theta'(t(r))}}\, e^{-i\nu(s-r)}\, ds\, dr, \tag{2}$$

where $\Psi_n^{\pm}(u, \nu) = \pm(u - t(u)\log n) - \nu u$ are the Riemann–Siegel phases. The off-diagonal $m \neq n$ cross terms in (2) encode the arithmetic pair-correlation structure of the integers through the irrationality of $\log m / \log n$.

2. The Half-Kernel

Definition 2.1 (Spectral factor / half-kernel)**.** The *half-kernel* $k \in L^2(\mathbb{R})$ of the zeta spectral process is the inverse Fourier transform of the amplitude $\sqrt{S(\nu)}$:

$$k(u) := \frac{1}{2\pi} \int_{-1}^{1} \sqrt{S(\nu)}\, e^{i\nu u}\, d\nu.$$

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It satisfies the factorization identity

$$(k * \tilde{k})(u) = C_h(u), \qquad \tilde{k}(u) := \overline{k(-u)},$$

where $C_h(\tau) = \mathbb{E}[h(u+\tau)\overline{h(u)}]$ is the autocovariance of h, and the moving-average representation

$$h(u) = \int_{\mathbb{R}} k(u-v)\, dW(v)$$

holds in $L^2(\Omega, \mathbb{P})$, where dW is the white-noise measure on $\mathbb{R}$.

Theorem 2.2 (Half-kernel as phase-flattened convolution)**.** *Let $\phi(\nu) = \arg \hat{h}(\nu)$ be the spectral phase of h, where $\hat{h}(\nu) = \lim_{T\to\infty} K_T(\nu)$. Define the* phase-flattening kernel

$$\kappa(\xi) := \frac{1}{2\pi} \int_{-1}^{1} e^{i\nu\xi}\, e^{-i\phi(\nu)}\, d\nu.$$

Then the half-kernel satisfies

$$k(u) = (h * \kappa)(u) = \int_{\mathbb{R}} h(s)\, \kappa(u-s)\, ds, \tag{3}$$

as an equality in $L^2(\mathbb{R})$. Equivalently,

$$k(u) = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \int_{\mathbb{R}} \frac{e^{i(s - t(s)\log n)} + e^{-i(s-t(s)\log n)}}{\sqrt{\theta'(t(s))}}\, \kappa(u-s)\, ds, \tag{4}$$

a superposition over all integers $n \geq 1$ of phase-flattened translates of the Riemann–Siegel amplitudes.

Proof. Write $\sqrt{S(\nu)} = |\hat{h}(\nu)| = \hat{h}(\nu)\, e^{-i\phi(\nu)}$. Then

$$k(u) = \frac{1}{2\pi} \int_{-1}^{1} \hat{h}(\nu)\, e^{-i\phi(\nu)}\, e^{i\nu u}\, d\nu.$$

Since $\hat{h} \in L^2([-1,1])$ by the Cramér isomorphism and $e^{-i\phi(\nu)}$ is bounded (of modulus 1), the product $\hat{h}(\nu)e^{-i\phi(\nu)}e^{i\nu u} \in L^1([-1,1])$. Substituting

$$\hat{h}(\nu) = \int_{\mathbb{R}} h(s)\, e^{-i\nu s}\, ds$$

(as an L^2 Fourier transform) and exchanging the ν-integral with the s-integral by the L^2 form of Fubini's theorem (justified because $\hat{h} \in L^2([-1,1])$ and the ν-domain is finite), we obtain

$$k(u) = \int_{\mathbb{R}} h(s) \left(\frac{1}{2\pi} \int_{-1}^{1} e^{i\nu(u-s)}\, e^{-i\phi(\nu)}\, d\nu \right) ds = (h * \kappa)(u).$$

Equation (4) follows by substituting the Riemann–Siegel expansion of h and using linearity of convolution. $\square$

Remark 2.3. The kernel κ is real and even because h is real-valued, which forces $\phi(-\nu) = -\phi(\nu)$ (odd spectral phase), making $\kappa(\xi) = \frac{1}{\pi}\int_0^1 \cos(\nu\xi - \phi(\nu))\,d\nu$ manifestly real. Consequently k is real, consistent with the half-kernel of a real-valued stationary process.

Remark 2.4 (Arithmetic content). Each term in (4) contributes oscillations at the Riemann–Siegel frequencies $\nu_n^{\pm} = \pm 1 \mp \log n/\theta'(t)$, which are dense in $[-1,1]$ as n and t vary. The phase-flattening kernel κ acts coherently across all n, stripping the phase while preserving the modulus $\sqrt{S(\nu)}$. The resulting k therefore encodes the full arithmetic of the integer pairs (m,n) through the off-diagonal terms of (2), in a single band-limited L^2 function supported spectrally on $[-1,1]$.

Remark 2.5 (Prolate connection). The half-kernel $k \in PW_{[-1,1]}$ (the Paley–Wiener space of functions band-limited to $[-1,1]$). The integral operator with kernel $C_h(u-v) = (k * \tilde{k})(u-v)$ restricted to a finite interval $[-T,T]$ is a finite-section Toeplitz operator whose eigenfunctions are prolate spheroidal wave functions with bandwidth parameter $c = T$. The weight $\sqrt{S(\nu)}$ on the spectral side specifies how the prolate eigenfunctions are weighted relative to the uniform (Lebesgue) measure on $[-1,1]$.